

VIDEO PROCESSING PIPELINE

Technical Documentation

Automated Institutional Media Processing Infrastructure

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Status: Operational Proof of Concept

GitHub: <https://github.com/GabrielDutraZ1/video-processing-pipeline-showcase.git>

EXECUTIVE SUMMARY

The project consists of an automated institutional media processing pipeline developed to organize, validate, transcode and monitor educational media continuously.

The infrastructure was built using Ubuntu Server, Bash scripting, FFmpeg, FFprobe, Systemd and VirtualBox operating in a fully headless environment.

The implemented architecture focuses on:

- automation;
- resilience;
- monitoring;
- validation;
- scalability;
- operational continuity.

The pipeline continuously monitors RAW media folders, automatically generates optimized outputs and validates transcoding integrity through automated recovery mechanisms.

The project demonstrates the feasibility of a resilient automated media processing infrastructure designed for scalable institutional environments.

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1. PROJECT OVERVIEW

The project was developed as a Proof of Concept (POC) for scalable institutional video processing infrastructure.

The primary goal was to create an automated environment capable of:

- organizing media;
- monitoring directories;
- processing videos;
- validating outputs;
- recovering corrupted files automatically.

The infrastructure simulates real-world institutional media processing scenarios focused on operational reliability and scalability.

2. PROJECT OBJECTIVES

Functional Objectives

- automate video transcoding;
- standardize media resolutions;
- reduce storage usage;
- validate transcoded outputs;
- organize mirrored folder structures;
- maintain continuous monitoring.

Infrastructure Objectives

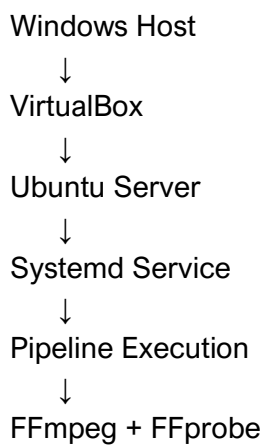
- implement headless execution;
- reduce manual intervention;

- provide persistent execution;
- create scalable architecture;
- simulate resilient infrastructure behavior.

3. INFRASTRUCTURE ARCHITECTURE

The environment operates through a virtualized Ubuntu Server infrastructure managed by VirtualBox.

Core Architecture

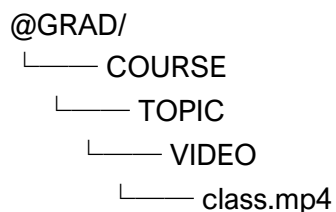


The processing service operates continuously in the background using Systemd persistence.

The infrastructure was intentionally designed to emulate a lightweight institutional processing environment with autonomous execution behavior.

4. FOLDER STRUCTURE

RAW Structure



Converted Structure

```
@CONVERTED_GRAD/  
├── COURSE  
│   ├── TOPIC  
│   │   ├── VIDEO  
│   │   │   ├── class_1080p.mp4  
│   │   │   ├── class_720p.mp4  
│   │   │   └── class_480p.mp4
```

The architecture maintains mirrored folder organization automatically.

This approach improves:

- scalability;
- organization;
- maintainability;
- institutional standardization.

5. PROCESSING WORKFLOW

The implemented workflow follows a continuous automated cycle:

```
RAW MEDIA  
↓  
MONITORING  
↓  
COPY VALIDATION  
↓  
TRANSCODING  
↓  
OUTPUT VALIDATION  
↓  
AUTO-HEALING  
↓  
ORGANIZED DELIVERY
```

Processing Stages

1. monitor RAW directory;
2. detect new media;

3. validate copy completion;
4. generate optimized outputs;
5. validate transcoded duration;
6. detect corrupted outputs;
7. automatically re-render invalid files;
8. restart monitoring cycle.

The monitoring cycle operates continuously in the background.

6. VALIDATION SYSTEM

The pipeline uses FFprobe validation routines to compare original and transcoded media duration.

Validation Logic

If duration mismatch exceeds tolerance:

- remove corrupted output
- wait filesystem unlock
- re-render media
- validate again

The validation mechanism significantly improves operational reliability by preventing corrupted outputs from remaining in the infrastructure.

The implemented logic allows continuous integrity checking during long processing cycles.

7. SELF-HEALING MECHANISM

One of the main architectural features implemented was the self-healing workflow.

The infrastructure was designed to autonomously recover from corrupted transcoding outputs without manual intervention.

Implemented Recovery Mechanisms

- retry logic;
- automatic corrupted file removal;

- re-renderization;
- continuous integrity verification;
- stabilization delays.

This creates resilient long-term operational behavior.

The retry architecture ensures that failed transcoding operations are automatically retried until validation succeeds.

8. LOGGING & MONITORING

Continuous monitoring was implemented through logging routines.

The pipeline continuously registers:

- transcoding events;
- monitoring cycles;
- validation operations;
- retry attempts;
- corruption detection;
- automatic recovery;
- operational status.

Screenshot Reference



```
gabriel@video-pipeline:~$ tail -f /mnt/scripts/logs/grad_pipeline.log
[14:19:55] 480p VALIDADO: RE0637_AISS_T2_V1_20261_480p.mp4
[14:20:02] 1080p VALIDADO: RE0637_AISS_T2_V2_20261_1080p.mp4
[14:20:03] 720p VALIDADO: RE0637_AISS_T2_V2_20261_720p.mp4
[14:20:04] 480p VALIDADO: RE0637_AISS_T2_V2_20261_480p.mp4
[14:20:10] 1080p VALIDADO: RE0637_AISS_T2_V3_20261_1080p.mp4
[14:20:11] 720p VALIDADO: RE0637_AISS_T2_V3_20261_720p.mp4
[14:20:12] 480p VALIDADO: RE0637_AISS_T2_V3_20261_480p.mp4
[14:20:19] 1080p CORROMPIDO - Recriando...
[14:20:19] Arquivo removido: RE0637_AISS_T2_V4_20261_1080p.mp4
[14:20:19] Convertendo 1080p: RE0637_AISS_T2_V4_20261.mp4
```

The monitoring layer provides operational transparency and easier troubleshooting.

The logging infrastructure allows continuous visibility into pipeline execution behavior.

9. HEADLESS INFRASTRUCTURE

The environment was configured for fully autonomous execution.

Implemented Headless Features

- automatic VM startup;
- automatic Ubuntu boot;
- automatic Systemd startup;
- background pipeline execution;
- continuous operation without GUI.

The environment operates continuously without graphical interaction requirements.

This significantly reduces operational overhead and manual interaction dependency.

The infrastructure simulates real-world automated server operation behavior.

10. PERFORMANCE RESULTS

Initial tests were intentionally performed using low-performance hardware under VirtualBox virtualization constraints.

Even under limited:

- CPU;
- RAM;
- storage I/O;

The infrastructure demonstrated:

- operational consistency;
- continuous execution;
- successful validation;
- automatic recovery behavior.

The architecture maintained stability during long processing cycles.

Even under constrained hardware conditions, the pipeline maintained operational consistency and recovery behavior.

This demonstrates strong infrastructure resilience.

11. STORAGE OPTIMIZATION ANALYSIS

The transcoding pipeline achieved highly significant storage optimization results.

Compression Efficiency

RESOLUTION	STORAGE REDUCTION
1080p	96%
720p	97%
480p	98%

These results demonstrate strong viability for:

- educational media archives;
- institutional storage optimization;
- large-scale media retention;
- scalable storage environments.

The compression efficiency achieved demonstrates strong viability for large-scale educational media archival systems.

The achieved storage reduction has significant implications for:

- storage cost reduction;
- media scalability;
- archival longevity;
- institutional optimization.

12. RELIABILITY ANALYSIS

The implemented infrastructure includes multiple reliability mechanisms.

Implemented Resilience Features

- continuous monitoring;
- retry logic;
- validation tolerance checks;
- automatic recovery;
- persistent service execution;
- filesystem stabilization handling;
- fault-tolerant workflow behavior.

These mechanisms help maintain operational continuity even during unstable conditions.

The architecture was intentionally designed to tolerate long-term autonomous execution.

13. SCALABILITY POTENTIAL

The architecture was intentionally designed as a scalable foundation for future enterprise-grade deployments.

Potential Future Improvements

- GPU transcoding;
- distributed processing nodes;
- AWS integration;
- Docker containerization;
- queue systems;
- dashboard monitoring;
- cloud storage integration;
- real-time metrics;
- distributed infrastructure scaling.

The modular workflow allows future expansion without requiring major architectural redesign.

The project architecture already demonstrates strong adaptability for cloud-native evolution.

14. TECHNOLOGIES USED

Technology	Purpose
Ubuntu Server	Processing Environment
Bash Script	Automation
FFmpeg	Video Transcoding
FFprobe	Validation
Systemd	Persistent Service
VirtualBox	Virtualized Infrastructure

The selected technologies were intentionally chosen to simulate realistic infrastructure automation environments.

15. TECHNICAL CHALLENGES

Several technical challenges were encountered during implementation.

Main Challenges

- filesystem locking;
- transcoding synchronization;
- validation timing;
- corruption detection;
- retry consistency;
- headless execution stability;
- virtualization resource limitations.

These challenges contributed significantly to the architectural evolution of the project.

The debugging and recovery mechanisms implemented became one of the strongest aspects of the infrastructure.

16. FUTURE IMPROVEMENTS

Planned Future Evolutions

- GPU accelerated transcoding;
- distributed transcoding nodes;
- queue-based processing;
- dashboard visualization;
- cloud-native deployment;
- AWS S3 integration;
- Kubernetes orchestration;
- Prometheus metrics;
- centralized monitoring.

These future implementations may transform the project into a large-scale institutional processing platform.

17. TECHNICAL SHOWCASE VALUE

The project demonstrates practical experience in:

- Linux infrastructure;
- Bash automation;
- resilient architectures;
- FFmpeg pipelines;
- validation systems;
- operational monitoring;
- headless virtualization;
- service orchestration;
- scalable infrastructure concepts.

The project represents a strong Proof of Concept for automated institutional media processing.

The infrastructure showcases both software automation and operational architecture concepts.

18. FINAL CONSIDERATIONS

The implemented infrastructure successfully demonstrated:

- automation viability;
- resilience mechanisms;
- operational consistency;
- scalable architecture potential;
- institutional applicability.

The project evolved from a simple transcoding script into a resilient automated processing architecture.

The implementation process also provided significant learning opportunities related to infrastructure design and fault tolerance.

19. CONCLUSION

The project successfully demonstrated the implementation of a resilient and scalable automated media processing infrastructure capable of operating continuously with minimal human intervention.

The implemented architecture combines:

- automation;
- validation;
- monitoring;
- recovery;
- organization;
- scalability.

The final result represents a strong technical showcase focused on operational resilience and institutional media processing automation.

The project architecture is prepared for future expansion into cloud and enterprise-grade infrastructures.

20. APPENDIX A — COMMANDS

Monitoring

```
tail -f /mnt/scripts/logs/grad_pipeline.log
```

Systemd Status

```
sudo systemctl status grad_pipeline
```

FFmpeg Process

```
ps aux | grep ffmpeg
```

RAW Structure

```
find /mnt/videos/@GRAD
```

Converted Structure

```
find /mnt/videos/@CONVERTED_GRAD
```

21. APPENDIX B — SCREENSHOT REFERENCES

Logs monitoring

```
gabriel@video-pipeline:~$ tail -f /mnt/scripts/logs/grad_pipeline.log
[14:19:55] 480p VALIDADO: RE0637_AISS_T2_V1_20261_480p.mp4
[14:20:02] 1080p VALIDADO: RE0637_AISS_T2_V2_20261_1080p.mp4
[14:20:03] 720p VALIDADO: RE0637_AISS_T2_V2_20261_720p.mp4
[14:20:04] 480p VALIDADO: RE0637_AISS_T2_V2_20261_480p.mp4
[14:20:10] 1080p VALIDADO: RE0637_AISS_T2_V3_20261_1080p.mp4
[14:20:11] 720p VALIDADO: RE0637_AISS_T2_V3_20261_720p.mp4
[14:20:12] 480p VALIDADO: RE0637_AISS_T2_V3_20261_480p.mp4
[14:20:19] 1080p CORROMPIDO - Recriando...
[14:20:19] Arquivo removido: RE0637_AISS_T2_V4_20261_1080p.mp4
[14:20:19] Convertendo 1080p: RE0637_AISS_T2_V4_20261.mp4
```

Systemd service

```
gabriel@video-pipeline:~$ sudo systemctl status grad_pipeline
[sudo: authenticate] Password:
• grad_pipeline.service - Pipeline Graduação
   Loaded: loaded (/etc/systemd/system/grad_pipeline.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-05-27 14:18:36 UTC; 5min ago
     Invocation: f6c74a59989647519cfe25505e784374
   Main PID: 1318 (grad_pipeline.s)
    Tasks: 29 (limit: 2696)
   Memory: 891.4M (peak: 891.4M)
      CPU: 14min 58.895s
   CGroup: /system.slice/grad_pipeline.service
           └─1318 /bin/bash /home/gabriel/grad_pipeline.sh
               └─2174 ffmpeg -nostdin -hide_banner -loglevel error -threads 1 -i /mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_04/RE0637_AISS_T2_V4_20261.mp4

May 27 14:19:56 video-pipeline grad_pipeline.sh[2062]: [14:19:55] 480p VALIDADO: RE0637_AISS_T2_V1_20261_480p.mp4
May 27 14:20:02 video-pipeline grad_pipeline.sh[2081]: [14:20:02] 1080p VALIDADO: RE0637_AISS_T2_V2_20261_1080p.mp4
May 27 14:20:03 video-pipeline grad_pipeline.sh[2091]: [14:20:03] 720p VALIDADO: RE0637_AISS_T2_V2_20261_720p.mp4
May 27 14:20:04 video-pipeline grad_pipeline.sh[2107]: [14:20:04] 480p VALIDADO: RE0637_AISS_T2_V2_20261_480p.mp4
May 27 14:20:10 video-pipeline grad_pipeline.sh[2125]: [14:20:10] 1080p VALIDADO: RE0637_AISS_T2_V3_20261_1080p.mp4
May 27 14:20:11 video-pipeline grad_pipeline.sh[2134]: [14:20:11] 720p VALIDADO: RE0637_AISS_T2_V3_20261_720p.mp4
May 27 14:20:12 video-pipeline grad_pipeline.sh[2144]: [14:20:12] 480p VALIDADO: RE0637_AISS_T2_V3_20261_480p.mp4
May 27 14:20:19 video-pipeline grad_pipeline.sh[2163]: [14:20:19] 1080p CORROMPIDO - Recriando...
May 27 14:20:19 video-pipeline grad_pipeline.sh[2168]: [14:20:19] Arquivo removido: RE0637_AISS_T2_V4_20261_1080p.mp4
May 27 14:20:19 video-pipeline grad_pipeline.sh[2172]: [14:20:19] Convertendo 1080p: RE0637_AISS_T2_V4_20261.mp4
lines 1-22/22 (END)
```

Raw structure

```
gabriel@video-pipeline:~$ find /mnt/videos/@GRAD
/mnt/videos/@GRAD
/mnt/videos/@GRAD/desktop.ini
/mnt/videos/@GRAD/RE0637_AISS_20261
/mnt/videos/@GRAD/RE0637_AISS_20261/APRESENTACAO_GERAL
/mnt/videos/@GRAD/RE0637_AISS_20261/APRESENTACAO_GERAL/RE0637_AISS_APG_20261.mp4
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_01
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_01/APRESENTACAO
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_01/APRESENTACAO/RE0637_AISS_T1_AP_20261.mp4
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_01
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_01/RE0637_AISS_T1_V1_20261.mp4
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_02
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_02/RE0637_AISS_T1_V2_20261.mp4
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_03
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_03/RE0637_AISS_T1_V3_20261.mp4
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_04
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_04/RE0637_AISS_T1_V4_20261.mp4
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_02
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_02/APRESENTACAO
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_02/APRESENTACAO/RE0637_AISS_T2_AP_20261.mp4
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_01
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_01/RE0637_AISS_T2_V1_20261.mp4
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_02
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_02/RE0637_AISS_T2_V2_20261.mp4
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_03
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_03/RE0637_AISS_T2_V3_20261.mp4
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_04
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_04/RE0637_AISS_T2_V4_20261.mp4
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_03
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_03/APRESENTACAO
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_03/APRESENTACAO/RE0637_AISS_T3_AP_20261.mp4
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_03/VIDEO_01
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_03/VIDEO_01/RE0637_AISS_T3_V1_20261.mp4
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_03/VIDEO_02
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_03/VIDEO_02/RE0637_AISS_T3_V2_20261.mp4
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_03/VIDEO_03
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_03/VIDEO_03/RE0637_AISS_T3_V3_20261.mp4
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_03/VIDEO_04
/mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_03/VIDEO_04/RE0637_AISS_T3_V4_20261.mp4
```

Converted structure

```

gabriel@video-pipeline:~$ find /mnt/videos/@CONVERTED_GRAD
/mnt/videos/@CONVERTED_GRAD
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/APRESENTACAO_GERAL
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/APRESENTACAO_GERAL/RE0637_AISS_APG_20261_1080p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/APRESENTACAO_GERAL/RE0637_AISS_APG_20261_480p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/APRESENTACAO_GERAL/RE0637_AISS_APG_20261_720p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_01
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_01/APRESENTACAO
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_01/APRESENTACAO/RE0637_AISS_T1_AP_20261_1080p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_01/APRESENTACAO/RE0637_AISS_T1_AP_20261_480p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_01/APRESENTACAO/RE0637_AISS_T1_AP_20261_720p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_01
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_01/RE0637_AISS_T1_V1_20261_1080p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_01/RE0637_AISS_T1_V1_20261_480p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_01/RE0637_AISS_T1_V1_20261_720p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_02
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_02/RE0637_AISS_T1_V2_20261_1080p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_02/RE0637_AISS_T1_V2_20261_480p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_02/RE0637_AISS_T1_V2_20261_720p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_03
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_03/RE0637_AISS_T1_V3_20261_1080p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_03/RE0637_AISS_T1_V3_20261_480p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_03/RE0637_AISS_T1_V3_20261_720p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_04
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_04/RE0637_AISS_T1_V4_20261_1080p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_04/RE0637_AISS_T1_V4_20261_480p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_01/VIDEO_04/RE0637_AISS_T1_V4_20261_720p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_02
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_02/APRESENTACAO
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_02/APRESENTACAO/RE0637_AISS_T2_AP_20261_1080p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_02/APRESENTACAO/RE0637_AISS_T2_AP_20261_480p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_02/APRESENTACAO/RE0637_AISS_T2_AP_20261_720p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_01
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_01/RE0637_AISS_T2_V1_20261_1080p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_01/RE0637_AISS_T2_V1_20261_480p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_01/RE0637_AISS_T2_V1_20261_720p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_02
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_02/RE0637_AISS_T2_V2_20261_1080p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_02/RE0637_AISS_T2_V2_20261_480p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_02/RE0637_AISS_T2_V2_20261_720p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_03
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_03/RE0637_AISS_T2_V3_20261_1080p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_03/RE0637_AISS_T2_V3_20261_480p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_03/RE0637_AISS_T2_V3_20261_720p.mp4
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_04
/mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_04/RE0637_AISS_T2_V4_20261_1080p.mp4

```

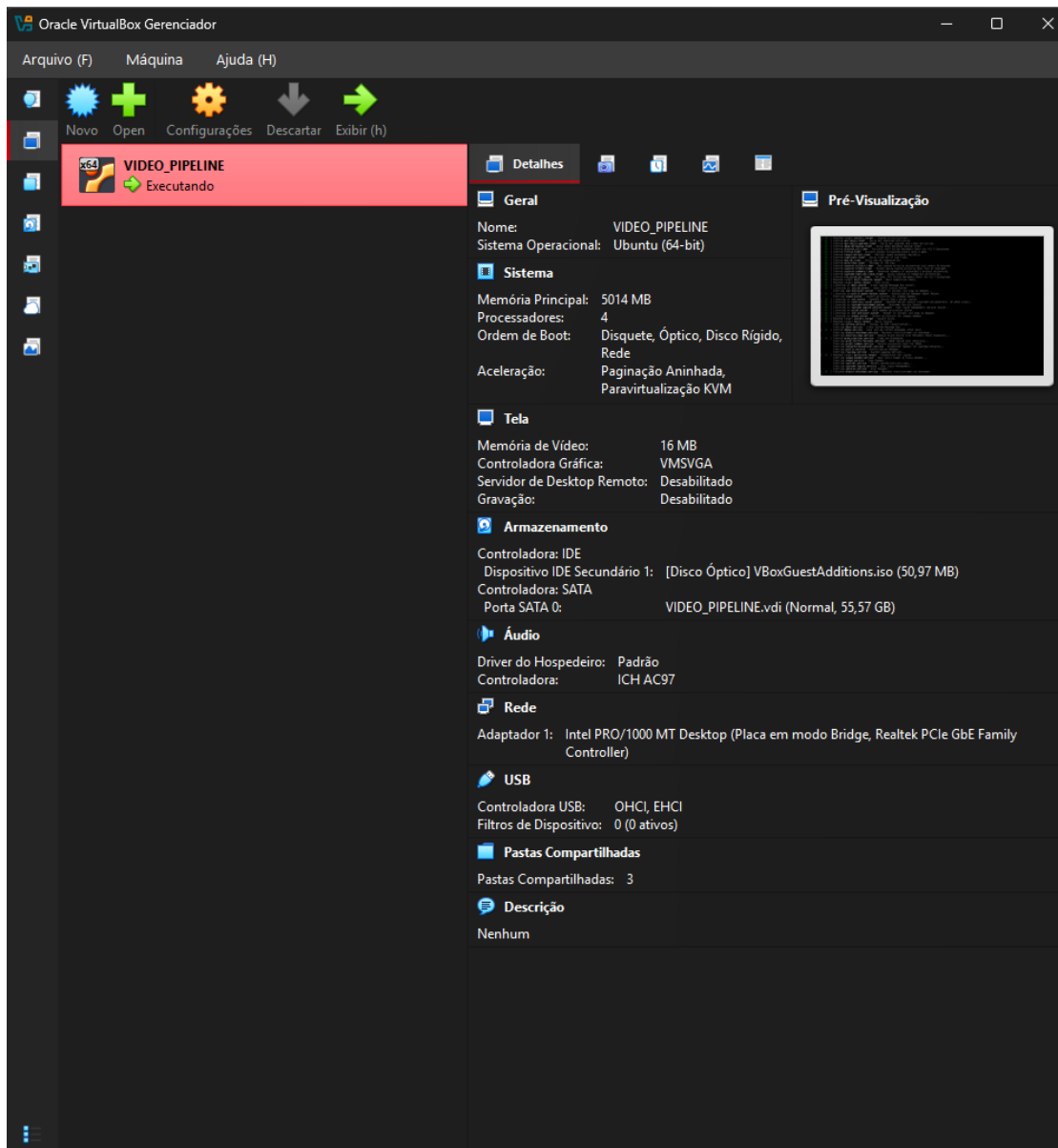
FFmpeg processing

```

gabriel@video-pipeline:~$ ps aux | grep ffmpeg
gabriel 2415 320 7.9 1327508 324404 ? Ssl 14:44 11:50 ffmpeg -nostdin -hide_banner -loglevel error -threads 1 -i /mnt/videos/@GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_04/RE0637_AISS_T2_V4_20261.mp4 -vf scale=2:720 -vcodec libx264 -crf 24 -preset fast -acodec aac -b:a 128k /mnt/videos/@CONVERTED_GRAD/RE0637_AISS_20261/TEMA_02/VIDEO_04/RE0637_AISS_T2_V4_20261_720p.mp4 -y
gabriel 2483 11.7 0.0 6716 2660 tty1 S+ 14:47 0:00 grep --color=auto ffmpeg
gabriel@video-pipeline:~$ _

```

Headless vm



These screenshots were collected during live infrastructure operation and demonstrate real execution behavior.

FINAL NOTE

Developed as a functional Proof of Concept focused on scalable institutional media processing automation.

The project demonstrates how lightweight automation and resilient infrastructure concepts can be combined to create efficient long-term operational workflows.